verification is straightforward—a bank teller can match a customer's face to their ID, and a security guard can check a badge against the person holding it. But in digital systems, this is much harder. The proximity problem—the fact that users authenticate remotely—means systems can't rely on physical presence.

When you hear the word *authentication*, you probably think of passwords. Passwords are as old as the first timeshare systems, when multiple people started sharing a computer. As the internet went mainstream and multi-tenanted websites became common, the use of passwords to authenticate service requestors became common. But passwords are just one way we can authenticate a request.

When Alice logs in to her employer's financial system, the system can't verify her identity through physical presence, so it relies on a digital ceremony—commonly called *logging in*—to recognize her. The ceremony starts with Alice presenting an identifier, laying claim to a particular account. The ceremony proceeds with Alice presenting one or more *authentication factors*—pieces of evidence that prove she is the legitimate account holder. Generally speaking, the more factors used, the stronger the authentication process. Note that the identifier isn't an authentication factor—it's a claim by the requester that needs to be verified.

Traditionally, authentication is categorized into three *primary factors*: knowledge, possession, and inherence. Modern authentication models also consider *contextual factors* that enhance security without requiring additional user input: behavior, location, and temporal.

3.5.1 Primary factors

The following sections describe the three primary authentication factors in detail, outlining their typical uses, strengths, and vulnerabilities.